

MATHEMATICAL ANALYSIS 1
HOMEWORK 2

- (1) Describe the following subsets of $\mathbb{R}$, specify what are their infimum and supremum, and determine whether their minimum and maximum are attained (explain your answers).
- (a) $A = \{x \in \mathbb{R} \mid (x+1)(x-1)(x-5) < 0\} \cap \{x \in \mathbb{R} \mid \frac{3x+1}{x-2} \geq 0\}$
 - (b) $B = \{x \in \mathbb{R} \mid x-4 \geq \sqrt{x^2-6x+5}\} \cup \{x \in \mathbb{R} \mid x+2 > \sqrt{x-1}\}$
 - (c) $C = \{x \in \mathbb{R} \mid x = \frac{1}{n-2}, n = 3, 4, 5, \dots\}$
- (2) Describe and sketch the following subsets of $\mathbb{R}^2$:
- (a) $A = \{(x, y) \in \mathbb{R}^2 \mid xy \geq 0\}$
 - (b) $B = \{(x, y) \in \mathbb{R}^2 \mid 1 + xy > 0\}$
 - (c) $C = \{(x, y) \in \mathbb{R}^2 \mid x^2 - y^2 > 0\}$
 - (d) $D = \{(x, y) \in \mathbb{R}^2 \mid x - y \neq 0\}$
 - (e) $E = \{(x, y) \in \mathbb{R}^2 \mid |x - y| < 2\}$
 - (f) $F = \{(x, y) \in \mathbb{R}^2 \mid |x - y| < -2\}$
- (3) Compute $\frac{100!}{98!}$. Explain your answer.
- (4) (a) Using the definition $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, prove Pascal's rule: for all integers n and k with $1 \leq k \leq n-1$,
- $$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$
- (b) Prove Newton's binomial formula. *Hint: prove by induction, using part (a).*
- (5) For each of the following sequences, determine whether there exists N such that the sequence is (monotonically) increasing or decreasing on $\{N, N+1, \dots\}$. If so, is it strictly increasing/decreasing there? Find the smallest such N . (Explain your answers.)
- (a) $a_n = \frac{n}{n+1}, n \in \mathbb{N}$
 - (b) $a_n = \frac{(-1)^n}{n}, n \in \mathbb{N}_+$
 - (c) $a_n = n^2 - 5n, n \in \mathbb{N}$
 - (d) $a_n = \frac{2^n}{n!}, n \in \mathbb{N}$
 - (e) The Fibonacci sequence, defined recursively by $a_0 = a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$. Proceed as follows:
 - (i) Prove by induction that for every $n \in \mathbb{N}$, both $a_n > 0$ and $a_{n+1} > 0$. (*Hint: take as the predicate $p(n) = 'a_n > 0 \text{ and } a_{n+1} > 0'$.*)
 - (ii) Show that $a_{n+1} - a_n = a_{n-1}$ for all $n \geq 1$, and use (i) to answer the question.
- (6) For the sequence $a_n = (-1)^n \frac{n}{n+1}, n \in \mathbb{N}$, find a subsequence that is increasing and a subsequence that is decreasing.
- (7) Describe the following sets: (explain your answers)
- (a) $\text{dom}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \frac{1}{\sin x}$.
 - (b) $\text{im}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \frac{1}{\sin x}$.
 - (c) $\text{dom}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \frac{1}{2+\sin x}$.
 - (d) $\text{dom}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = 10^x$.
 - (e) $\text{im}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = 10^x$.
 - (f) $\text{dom}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \log_{10}(x)$.
 - (g) $\text{im}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \log_{10}(x)$.
- (8) For the following functions f and subsets $B \subseteq \mathbb{R}$, describe $f^{-1}(B)$.
- (a) $f: \{3, 4, 5, \dots\} \rightarrow \mathbb{R}$ defined by $f(n) = \frac{1}{n-2}$, and $B = (0, 1)$.

- (b) In the previous question, what if $B = [0, 1]$?
 - (c) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^4$, $B = [1, 16]$.
 - (d) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3 + 1$, $B = (-26, -7]$.
 - (e) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$, $B = \{0\}$.
- (9) Let $f : X \rightarrow Y$ be a function between two sets X and Y .
- (a) Prove that $A \subseteq f^{-1}(f(A))$ for any $A \subseteq X$.
 - (b) Give an example of when $A \neq f^{-1}(f(A))$.